

Hemant Pathak

+91-9548741368 Hemantpathak1411@gmail.com
[LinkedIn-hemant-pathak1411](#) [GitHub-HEMANT2027](#)



EDUCATION

Indian Institute of Technology (ISM) Dhanbad

Integrated M.Tech in Mathematics and Computing (GPA: 8.90/ 10.00)

Expected May 2028

Dhanbad, Jharkhand

- **Relevant Coursework:** Data Structures & Algorithms, Operating Systems, Database Management Systems, Object-Oriented Programming

EXPERIENCE

Apparent Energy Solutions Pvt. Ltd.

Jan 2026 – Present

Software Engineering Intern

Remote

- Built the core request pipeline for a production assistant on a utility meter-data platform — identity-header auth, per-user rate limiting via a **Postgres token bucket**, and streaming responses to the UI over SSE using **FastAPI**.
- Designed a crash-safe per-turn **orchestrator** that persists every message and tool-call row to Postgres during the loop, so a mid-turn failure leaves a complete audit trail rather than partial state.
- Implemented an observability layer subscribing to the event stream, recording token usage, tool latency, and finish reason per turn.
- Integrated a text-to-SQL service over meter/interval data with curated schema and 50+ example queries to eliminate hallucinated table/column references.

Aeroqube Inc.

Sep 2025 – Dec 2025

Software Engineering Intern [\[Certificate\]](#)

Remote

- Built multimodal reasoning agents with **LangChain/LangGraph** and a FAISS-backed retrieval pipeline serving context-aware summaries over unstructured data.
- Built preprocessing pipelines for CT / X-ray data; validated segmentation quality against held-out scans.
- Deployed inference on **AWS SageMaker** with model quantization.

PROJECTS

Shell.ai Fuel Blend AI Challenge 2025

Rank 86 / 7,020 (Team: “Here it is”) [\[Leaderboard\]](#)

- Built a multi-target regression pipeline predicting 10 fuel-blend properties from 55 input variables; engineered interaction/heterogeneity features for non-linear mixing behavior.
- Benchmarked LightGBM, CatBoost, XGBoost, Random Forest, and linear models with 5-fold CV; built hybrid ensembles via prediction blending and target-level optimization.

IntelliDesk AI | Python, Pinecone, SentenceTransformers, Groq

- Built an automated support system: incoming emails classified by category/priority, then answered via LLM — eliminating manual triage.
- Implemented semantic retrieval with **Pinecone + SentenceTransformers**; deterministic structured-JSON inference via **Groq** (LLaMA-3.1-8B).

CODING PROFILES

Codeforces: [HemantPathak](#) — 657 problems solved (rating: Pupil / 1282)

LeetCode: [Hemant1411](#) — 258 problems solved (contest rating: 1624)

TECHNICAL SKILLS

Languages: C++, Python, SQL (MySQL), C

Backend: FastAPI, PostgreSQL

Tools / Cloud: Git, GitHub, AWS SageMaker, Linux

ACHIEVEMENTS

- **Runner-up, AlgoVerse Algorithmic Trading Hackathon:** Backtested a quantitative signal-generation strategy, outperforming 50+ teams before an industry panel.

POSITIONS OF RESPONSIBILITY

Student Placement Coordinator: Career Development Centre, IIT (ISM) Dhanbad (Session 2025–2026)

Member: Fintech Club, IIT (ISM) Dhanbad